

Force-Directed Scheduling for High-Level Synthesis

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Algorithm 1 Force-Directed Scheduling

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1: compute ASAP and ALAP under latency  $L$ 
2: while unscheduled operations remain do
3:   build  $q_k$  from current time frames {Eq. (2)}
4:   for all unscheduled  $i$  and step  $l \in [t_i^S, t_i^L]$  do
5:     compute  $F_i(l)$  {Eq. (3)}
6:   end for
7:    $(i^*, l^*) \leftarrow \arg \min F_i(l)$ 
8:   fix  $i^*$  at  $l^*$ ; tighten frames of its predecessors/successors
9: end while
10: return schedule
    
```

Abstract—

*Index Terms—*high-level synthesis, scheduling, force-directed scheduling, resource sharing, data-flow graph

I. INTRODUCTION

II. BACKGROUND AND MATHEMATICAL MODEL

The time frame of operation v_i is $[t_i^S, t_i^L]$, where t_i^S and t_i^L are its ASAP and ALAP start steps; the frame width is $w_i = t_i^L - t_i^S + 1$.

$$p_i(l) = \begin{cases} \frac{1}{w_i}, & t_i^S \leq l \leq t_i^L \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

III. FORCE-DIRECTED SCHEDULING

For every resource type k , the distribution graph is

$$q_k(l) = \sum_{i: \text{type}(i)=k} p_i(l). \quad (2)$$

When operation i is tentatively assigned to step l , its probability changes by $\Delta p_i(m) = [m = l] - p_i(m)$ over its frame. The resulting force is

$$F_i(l) = \underbrace{\sum_{m=t_i^S}^{t_i^L} q_k(m) \Delta p_i(m)}_{\text{self-force}} + \sum_{j \in \text{pred} \cup \text{succ}} F_j^{\text{ind}}(l), \quad (3)$$

where each indirect term F_j^{ind} has the same form, evaluated over the (shifted) frame of a dependent operation whose mobility is reduced by the assignment.

TABLE I

ADD DISTRIBUTION GRAPH (EQ. (2)) BEFORE AND AFTER SCHEDULING

step	1	2	3	4
initial q_{ADD}	0.00	1.00	1.50	0.50
final q_{ADD}	0.00	1.00	1.00	1.00

TABLE II

FUNCTIONAL UNITS REQUIRED AT LATENCY $L = 4$

Resource	ASAP baseline	FDS
Multipliers	3	2
Adders	2	1

IV. COMPLEXITY AND OVERHEAD

V. IMPLEMENTATION

Listing 1. Main feature: total force and minimum-force selection.

```

1 double force(int i, int l, const vector<int>& pin,
2             const vector<int>& asap, const vector<int>& alap,
3             const vector<vector<double>>& q) {
4     vector<int> pin2 = pin; pin2[i] = 1; // tentatively pin i
5     // at l
6     vector<int> a2, b2; frames(pin2, a2, b2); // propagate to
7     // dependents
8     double F = 0.0;
9     for (int o = 0; o < N; ++o) {
10        if (pin[o] >= 0) continue; // committed -> unchanged
11        int t = ops[o].type;
12        double pOld = 1.0 / (alap[o] - asap[o] + 1);
13        double pNew = 1.0 / (b2[o] - a2[o] + 1);
14        for (int s = 1; s <= L; ++s) {
15            double oP = (s >= asap[o] && s <= alap[o]) ? pOld :
16                0.0;
17            double nP = (s >= a2[o] && s <= b2[o]) ? pNew : 0.0;
18            if (oP != nP) F += q[t][s] * (nP - oP);
19        }
20    }
21    return F; // lower force = better
22 }
23 // selection inside the scheduling loop:
24 // for each unscheduled i, each step l in frame: f = force
25 // (...);
26 // keep (i*, l*) with the minimum f; then fix it and repeat.
    
```

VI. APPLICATION EXAMPLE AND RESULTS

VII. DISCUSSION

A. *Context and Trade-offs*

B. *Critical Evaluation*

VIII. CONCLUSION

STATEMENT ON THE USE OF AI TOOLS

REFERENCES

- [1] P. G. Paulin and J. P. Knight, "Force-directed scheduling for the behavioral synthesis of ASICs," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 8, no. 6, pp. 661–679, Jun. 1989.
- [2] G. De Micheli, *Synthesis and Optimization of Digital Circuits*. New York, NY, USA: McGraw-Hill, 1994.
- [3] J. Teich, *Digitale Hardware/Software-Systeme: Synthese und Optimierung*. Berlin, Heidelberg, Germany: Springer, 1997.